Experiment 3 — Power Transmission Mechanism

Design of Mechanisms (BE05000341) · Quiz — Short Answers

Q1. Standard systems of gear teeth; 14.5° and 20° involute systems.

A standard system fixes the pressure angle, addendum, dedendum and clearance so that gears from different makers mesh. **14.5° full-depth** gives smoother/quieter meshing and smaller radial force but has a weaker, narrower tooth and needs ≥ 32 teeth to avoid undercut. **20° full-depth** has a wider, stronger tooth, needs only ≥ 18 teeth and resists wear better; the radial force is larger but it is now the most used standard.

Q2. Law of gearing.

Ans: For a constant velocity ratio, the common normal at the point of contact of the two mating teeth must always pass through a fixed point (the pitch point P) on the line of centres.

Q3. Beam strength of a spur gear (Lewis equation).

Treating a gear tooth as a cantilever loaded at the tip, the Lewis equation gives the tangential load the tooth can carry before it yields in bending.

Ans: **Sb = m · b · σb · Y** (N), where **Y = π y**; for 20° full-depth: **y = 0.154 − 0.912 / z**; for 14.5°: **y = 0.124 − 0.684 / z**.

Q4. Parameters for selecting rolling-contact bearings.

Select by: **type of load** (radial / thrust / combined), **magnitude and direction**, **shaft speed**, **expected life (L10)**, **shaft diameter**, **mounting space**, **shock and vibration**, **required accuracy and stiffness**, **lubrication and operating temperature**, **sealing / environment**, and **cost / availability**.

Q5. Important bearing terms.

Static capacity C_0 — load that produces a permanent indentation 0.0001× rolling-element diameter. **Dynamic capacity C** — constant radial load that gives a basic rating life L10 of 1×10^6 revolutions. **Rating life (L10)** — life reached or exceeded by 90 % of a group of identical bearings. **Equivalent load Pe = XVFr + YFa** — the single radial load with the same effect as the actual combined load.

Q6. Typical causes of failure in rolling-contact bearings.

Fatigue spalling (normal end-of-life — flakes break out of the race), **wear/abrasion** by dirt, **corrosion** from water/moisture, **brinelling** from static shock loads, **false brinelling** from vibration at rest, **electric arcing**, **overheating** from poor lubrication, **misalignment**, and **wrong fit/mounting**.

Q7. Performance of a hydrodynamic bearing.

A journal rotating in a bearing drags lubricant into the clearance; a wedge-shaped film builds pressure and lifts the shaft off the metal. Performance (minimum film thickness, friction, flow, temperature rise, power loss) depends on the **Sommerfeld number** $S = (r/c)^2 \cdot \eta N/p$, and is read from the McKee / Raimondi–Boyd charts. Design aim: **h_0 > safe limit**, **temperature within limits**, **minimum power loss**.

Q8. Journal (sliding) vs anti-friction (rolling) bearings.

Point	Sliding (Journal)	Rolling (Anti-friction)
Friction	Depends on film; low only after start-up	Low from start, almost constant
Starting torque	High	Low
Shock / noise	Takes shock well; silent	Noisier; shock-sensitive
Space (radial)	Smaller	Larger
Cost	Low; easy to make	Higher; standard parts
Life	Very long with good lubrication	Fatigue-limited (L10)

Q9. Numerical — Spur gear pair for a pump drive (power 2.5 kW, pinion 1440 rpm, gear 864 rpm, 20° full-depth, FOS = 1.5).

Speed ratio i = 1440/864 = 1.67. Choose pinion teeth zp = 21 → gear teeth zg = 35. Try module **m = 3 mm**, face width **b = 30 mm (= 10 m)**. Pitch dias dp = 63 mm, dg = 105 mm → centre distance **84 mm**. Tangential force Ft = 2T/dp with T = 60 × 2500/(2π·1440) ≈ 16.6 N·m → Ft ≈ 527 N; design load Fd = Ft × FOS × Cv ≈ **3335 N**. Beam strength Sb = m b σb Y with σb = 236 N/mm², Y = π(0.154 − 0.912/21) = 0.349 → **Sb ≈ 3752 N**. Wear strength Sw = dp b Q K ≈ **3402 N**. Both > 3335 N → design safe.

Ans: zp = 21, zg = 35; **m = 3 mm, b = 30 mm**; centre dist **84 mm**; Sb = 3752 N, Sw = 3402 N — both > Fd = 3335 N (safe).

Q10. Numerical — Beam strength, wear strength and rated power of a given gear pair (zp = 20, zg = 50, m = 5, b = 50, dp = 100).

Y_p = π (0.154 − 0.912/20) = 0.340. Sb = m b σb Y = 5 × 50 × 231.5 × 0.340 ≈ **19 680 N (19.7 kN)**. Wear strength Sw = dp b Q K with Q = 2zg/(zp+zg) = 1.43 and K ≈ 1.6 → **Sw ≈ 11.52 kN** (limiting). Rated tangential load (effective) Feff = min(Sb, Sw)/FOS ≈ 11 520/1.5 ≈ 7680 N → Rated power P = Feff × vp = 7680 × (π × 0.100 × 1440/60) ≈ **8.24 kW** (limited by wear).

Ans: Sb = **19.68 kN**; Sw = **11.52 kN** (limiting); rated power ≈ **8.24 kW**.

Q11. Numerical — Dynamic load capacity of a ball bearing under a work cycle.

For each stage, equivalent load Pe = XVFr + YFa and life fraction. Combine with Pe_mean = (Σ Pe³·N·t)^(1/3)/(ΣN·t)^(1/3) ≈ **3.61 kN**. For required life L10 (in millions of revs), **C = Pe (L10)^(1/3) ≈ 33.2 kN**. Pick a bearing with catalogue C ≥ 33.2 kN.

Ans: Equivalent load **Pe ≈ 3.61 kN**; required dynamic capacity **C ≈ 33.2 kN**.

Q12. Numerical — Hydrodynamic journal bearing (d = 50 mm, l = 50 mm, N = 1200 rpm, W = 4 kN, c = 50 µm, SAE 20 at 70 °C → η = 20 cP).

Pressure p = W/(l·d) = 4000/(50·50) = 1.6 N/mm² = 1.6 MPa. Sommerfeld S = (r/c)² · η N / p = (25/0.05)² · (0.020)(20)/(1.6×10⁶) = **0.125**. Chart gives $h_0/c \approx 0.44$ → **h_0 ≈ 22 µm**, coefficient of friction (r/c) f ≈ 1.8 → **f ≈ 0.0037**. Friction power = f · W · v with v = π d N/60 = 3.14 m/s → ≈ **0.87 kW**.

Ans: (a) **h_0 ≈ 0.022 mm**; (b) **f ≈ 0.0037**; (c) power lost ≈ **0.87 kW**.